VEPOLINK
MONITORING PURITY

ENVIRONMENTAL INTELLIGENCE × GAS SAFETY

FIXED GAS DETECTION · HAZARDOUS AREAS

Fixed Gas Detector
Always on watch

A flameproof, wall-mounted transmitter that monitors toxic and flammable gas around the clock — feeding 4–20 mA and RS-485 to your control panel so a leak becomes an alarm, and an alarm becomes a shutdown.



24 / 7
CONTINUOUS MONITORING

4–20 mA
LOOP-POWERED OUTPUT

20+ gases
SENSOR OPTIONS

EX D IIC T6 FLAMEPROOF | **IP65** DUST & WATER | **RS-485** · RELAYS

30s	2% FS	1% FS/yr	1MB
RESPONSE TIME	PRECISION	ZERO DRIFT	NON-VOLATILE DATA LOGGER

01 OVERVIEW

Protection that *never clocks off.*

Toxic and flammable gases are part of everyday process work, and wherever the "EX" mark is posted a leak is a fire or a poisoning waiting for a spark or a breath. A fixed detector installed at the high-priority points — compressor houses, tank farms, loading bays, sewer wells — watches those points 24 hours a day and raises the alarm before it is too late.

The Vepolink fixed detector is a flameproof transmitter with a high-brightness LED display and a plug-in sensor head, working on electrochemical and catalytic principles. It runs on 24 V DC and returns a standard 4–20 mA signal, with RS-485, 0–5 V and relay outputs, to a control panel, PLC or SCADA that displays the concentration and triggers alarms and shutdowns.

AT A GLANCE

- ELECTROCHEMICAL
- CATALYTIC BEAD
- 4-20 MA
- RS-485 · 0-5 V
- RELAY OUTPUTS
- WIRELESS OPTION
- EX D IIC T6
- IP65

· PRODUCT FEATURES

Flameproof Housing

Ex d IIC T6 enclosure contains any internal arc, so the detector itself can never be the ignition source in the area it protects. IP65 against dust and water.

Standard Loop Output

Two-wire 4–20 mA plus RS-485 and relays talk to any control panel, PLC or DCS — no proprietary controller required.

Readable at a Distance

A high-brightness LED display shows live concentration on the unit itself, so a technician can check a point without a laptop or a panel.

One Body, Twenty Gases

The same transmitter accepts sensor heads for more than twenty toxic and flammable gases, so a site standardises on one part and one spares kit.

02 DETECTION PRINCIPLE

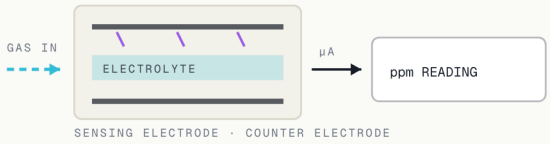
Sense, transmit, *act.*

The system supplies 24 V DC to the detector and reads back a standard 4–20 mA signal. Everything between the sensor and the shutdown relay is designed around that loop.

ELECTROCHEMICAL

Toxic Gases & Oxygen

CO · H_2S · SO_2 · NH_3 · CL_2 · O_2



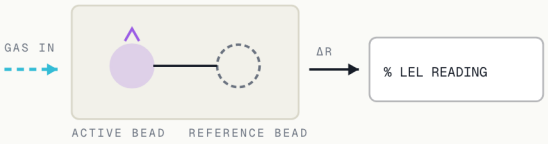
Gas diffusing into the cell is oxidised or reduced at the sensing electrode, producing a current in direct proportion to concentration. Low power, ppm resolution, and selective by electrode chemistry.

`Current ∝ concentration`

CATALYTIC BEAD

Combustible Gases

CH_4 · C_3H_8 · C_2H_2 · H_2 · 0–100 % LEL



Flammable gas burns on a heated catalytic bead, raising its resistance relative to an inert reference bead. The imbalance reads out as percent of lower explosive limit.

`Reads in % LEL`

SIGNAL PATH

Loop to Control Panel

24 V DC IN · 4-20 MA · RS-485 · RELAYS



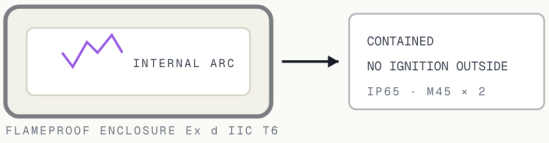
The transmitter converts sensor output to a 4–20 mA current the panel reads directly. RS-485 carries digital values and status; relays drive sirens, beacons and process shutdown.

`4 mA = zero · 20 mA = full scale`

PROTECTION

Flameproof by Design

EX D IIC T6 · IP65 · M45 × 2 THREAD



A flameproof enclosure withstands an internal explosion and prevents its transmission to the surrounding atmosphere — the requirement for Zone 1 installations in the most sensitive gas group.

`Safe in the area it protects`

03 GAS RANGES

Ranges and *alarm points.*

Reference settings for the most common sensor heads. Alarm points are factory defaults — users set them to suit their own site conditions and exposure limits.

GAS	NAME	RANGE · RES.	ALARM
CH_4	Methane	0–100 · 1 % LEL	25 % LEL
C_3H_8	Propane	0–100 · 1 % LEL	25 % LEL
C_2H_2	Acetylene	0–100 · 1 % LEL	25 % LEL
CO	Carbon Monoxide	0–1000 · 1 ppm	50 ppm
H_2S	Hydrogen Sulphide	0–100 · 1 ppm	10 ppm
H_2	Hydrogen	0–1000 · 1 ppm	35 ppm
SO_2	Sulphur Dioxide	0–20 · 1 ppm	5 ppm
NH_3	Ammonia	0–100 · 1 ppm	35 ppm

GAS	NAME	RANGE · RES.	ALARM
NO	Nitric Oxide	0–250 · 1 ppm	35 ppm
NO_2	Nitrogen Dioxide	0–20 · 1 ppm	10 ppm
Cl_2	Chlorine	0–20 · 1 ppm	2 ppm
O_3	Ozone	0–20 · 1 ppm	2 ppm
ETO	Ethylene Oxide	0–20 · 1 ppm	10 ppm
CO_2	Carbon Dioxide	0–5000 · 1 ppm	1000 ppm
HCN	Hydrogen Cyanide	0–50 · 1 ppm	10 ppm
O_2	Oxygen	0–30 · 0.1 % vol	18 % / 23 %

STEP 01

Mount

Wall, pipeline or flow-through fitting at the release point, sensor head down, with the M45 × 2 thread and G1/2 gland.

STEP 02

Wire

24 V DC supply and 4–20 mA loop to the panel; RS-485 for digital readout, relays to siren and shutdown.

STEP 03

Set alarms

Accept the reference alarm points or set your own per gas from the panel or the unit.

STEP 04

Monitor

Concentration is displayed locally and at the panel; the 1 MB logger keeps the record.

Further sensor heads — formaldehyde, HCl, NOx, N_2, phosphine, hydrogen fluoride and others — are available to order. Alarm points shown are for reference only and should be reviewed against site exposure limits.

04 SPECIFICATIONS

Technical *specifications.*

MEASUREMENT

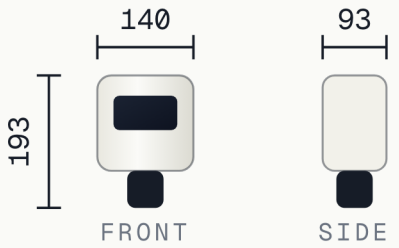
Gas Detection	O_2, O_3, CO, CO_2, CH_4, Cl_2, CH_2O, HCl, H_2, H_2S, HCN, NO, NO_2, NOx, NH_3, N_2, SO_2, ETO, PH_3, HF
Detection Principle	Electrochemical & catalytic
Sampling Method	Wall · pipeline · flow-through
Resolution	Down to 0.001 ppm, sensor dependent
Precision	2 % FS
Repeatability	1 %
Linearity Error	1 %
Zero Drift	1 % FS per year
Response Time	30 s
Recovery Time	10 s

OUTPUT · BUILD · RATING

Output Signal	4-20 mA · 0-5 V · RS-485 · relays
Wireless	Digital wireless, optional
Computer Interface	RS-232
Data Logger	1 MB non-volatile memory
Display	High-brightness LED
Working Voltage	24 V DC
Connection	G1/2 male, flameproof hose pipe · M45 × 2 mm thread
Explosion Proof	Ex d IIC T6
Ingress Protection	IP65
Working Temperature	10 - 40 °C

INSTRUMENT DIMENSIONS

193 × 140 × 93 MM · FRONT & SIDE ELEVATION



THREAD
M45 × 2 mm

05 INSTALLATION & UPKEEP

Mounted once, *trusted for years.*

> Placement decides performance. Heavier-than-air gases such as propane and H_2S are detected **low, near floor level**; lighter gases such as methane and hydrogen **high, near the ceiling**; toxic gases at **breathing height** where people work. Every point is then **bump tested and calibrated on a schedule** with certified gas.

01 · LOCATION

At the release point

Close to valves, flanges, compressors and vessels where a leak would originate, considering airflow and gas density.

02 · MOUNTING

Wall, pipe or duct

Wall bracket, pipeline fitting or flow-through cell. Sensor head points down so dust and water do not collect on the grille.

03 · WIRING

Flameproof gland

24 V DC and signal through the G1/2 certified gland; screened cable for the 4–20 mA loop and RS-485 bus.

04 · PANEL

Display and act

The control panel shows every point, drives sirens and beacons, and can trip ventilation or process shutdown via relays.

05 · CALIBRATION

Certified gas, on schedule

Zero in clean air, span with a known concentration through a calibration cap — without opening the flameproof housing.

06 · SENSOR CHANGE

Plug-in head

When response slows, the sensor head swaps out and the point is recalibrated. The transmitter stays on the wall.

FIXED AND PORTABLE TOGETHER

Fixed detectors guard the places a leak is most likely; they cannot follow a technician into a vessel or under a floor. Most sites pair them with Vepolink portable single- and multi-gas detectors, so the plant is watched continuously and every person is watched individually.

SCALING A SYSTEM

From a single point on one 4–20 mA loop to dozens of points on an RS-485 bus, the same transmitter scales with the site. Add points as the plant grows; the panel, the spares and the calibration routine stay the same.

06 APPLICATIONS

Where it *works.*

Anywhere a gas is stored, moved or made — and anywhere the "EX" mark says a leak must never meet a spark.

Refineries & Petrochemicals	Chemical & Fertiliser Plants	LPG & Gas Storage
Sewage & Water Treatment	Power & Boiler Houses	Steel & Metallurgy
Pharma & Sterilisation (ETO)	Cold Storage & Ammonia Plants	Battery Rooms & Hydrogen

PERSONNEL AND ASSET PROTECTION

A fixed detector serves two purposes at once: it warns the people in the area to leave, and it gives the control system the signal to close a valve, start a fan or stop a process. The first protects lives; the second protects the plant and the neighbourhood around it.

CHOOSING THE SENSOR

Catalytic LEL heads for methane, propane and hydrogen where fire is the risk; electrochemical heads for CO, H_2S, chlorine, ammonia and other toxics where exposure is; oxygen heads wherever inert gas or displacement could leave a space unbreathable. Tell us the process and we will map the points and the sensors.

07 FAQ

Frequently asked *questions.*

What is the fixed gas detector?

A continuous, permanently installed gas detection transmitter for industrial safety that monitors toxic and flammable gases 24 hours a day and reports to a control panel.

Which gases can it monitor?

A wide range including CH_4, CO, H_2S, SO_2, NH_3, NOx, Cl_2, O_3, HCN, CO_2, ETO and oxygen — more than twenty sensor heads on the same transmitter body.

Does it support alarms and alerts?

Yes. Audible and visual alarms on the panel, relay outputs for sirens and shutdown, and a local LED display on the unit, all triggered when concentration passes the set limit.

How is it connected and powered?

It runs on 24 V DC and returns standard 4–20 mA and RS-485 signals, with 0–5 V and relay outputs, to interface with control panels, PLC and SCADA.

Is it suitable for hazardous areas?

Yes. The Ex d IIC T6 flameproof housing and IP65 rating suit Zone 1 installations across industrial leak detection, workplace safety and hazardous-area monitoring.

Can the alarm points be changed?

Yes. The factory alarm points are reference values; users set their own per gas to suit site conditions and exposure limits.

How is it mounted?

Wall, pipeline or flow-through, using the M45 × 2 mm thread and a G1/2 flameproof gland. Location follows the density of the target gas.

How is it calibrated?

Zero in clean air and span with certified gas through a calibration cap, on a schedule — without opening the flameproof housing.

Does it log data?

Yes. A 1 MB non-volatile logger holds the record, retrievable over the RS-232 computer interface or through the panel.

Can it work with portable detectors?

Yes, and it should. Fixed points guard the plant continuously; Vepolink portable single- and multi-gas detectors guard the people wherever they go.

This brochure is general product information. Configuration, dimensions and performance figures are indicative and confirmed at order. Gas detectors are safety equipment: follow the user manual, site procedures and applicable standards for siting, calibration and use. Specifications may change without notice as the product is improved.

VEPOLINK
MONITORING PURITY

EVERY POINT, EVERY HOUR

Catch the leak *before it counts.*

Tell us the gases, the areas and the panel you report to, and we'll map the detection points, sensors, alarm levels and wiring as one system. Pick how you'd like to start.

01 **Talk to our engineers**
Speak directly to the team that sites, installs and calibrates fixed gas detection. No call centre.
+91 99538 86178 →

02 **Request a detection layout** RECOMMENDED
Send your plant drawing and gas list — we'll spec points, sensor heads, alarm levels, panel and wiring.
sales@vepolink.com →

03 **Book a demonstration**
See a live detector on a panel — alarm, relay trip and calibration — with our engineering team.
Request a demo →

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